An online adaptive sampling algorithm for nonsmooth stochastic optimization with difference-of-convex objectives *

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Abstract

We propose an adaptive online algorithm for stochastic nonsmooth nonconvex optimization problems with difference-of-convex (DC) objectives that achieves state-of-art sample complexity. The sample size and step size are determined by the current estimates of stochastic quantities. The Sample Average Approximation (SAA) error for the linearized concave parts is closely related to the convergence of the SAA for subdifferential mappings. We derive a novel $O(\sqrt{p/n})$ point-wise rate (modulo logarithmic factors) where p is the dimension and n is the sample size. Our result complements the state-of-art $O(\sqrt[4]{p/n})$ result, almost matching the same SAA convergence rate in the smooth case. Due to the different convergence rates and DCA's special structure, we should use different sampling sizes to achieve the optimal error-sample size trade-off. Numerical experiments show that our adaptive strategy improves the algorithm's performance and reduces its worst-case sampling size.

Keywords: stochastic, nonsmooth, sampling, adaptive, difference-of-convex, sample average approximation, subdifferential mapping

1 Introduction

Consider the class of nonsmooth nonconvex optimization problems in the form of

$$\underset{x\in C}{\text{minimize}} \quad f(x),$$

where $C \subset \mathbb{R}^p$ is a convex compact set. The nonsmooth stochastic objective function $f : \mathbb{R}^p \to \mathbb{R}$ is given as $f(x) = \mathbb{E}_{\xi,\zeta}[F(x,\xi,\zeta)]$, where the expectation $\mathbb{E}_{\xi,\zeta}$ is the expected value of the function $F(x,\xi,\zeta)$ with respect to ξ and ζ. The random vector ξ and ζ is defined over probability spaces $(\Omega_1, \mathcal{F}_1, \mathbb{P}_1)$ and $(\Omega_2, \mathcal{F}_2, \mathbb{P}_2)$, respectively. In this work, we focus on $F : (\mathbb{R}^p, \Omega_1, \Omega_2) \to \mathbb{R}$, which are difference-of-convex functions with the decomposition

$$F(x,\xi,\zeta) = G(x,\xi) - H(x,\zeta),$$

where $G(x,\xi)$ and $H(x,\zeta)$ are Carathéodorian, i.e., it is continuous in $x \in C$ for all $\xi \in \Omega_1$, $\zeta \in \Omega_2$ and Borel measurable in ξ and ζ for all $x \in C$.

Let $g = \mathbb{E}_\xi[G(x,\xi)]$ and $h = \mathbb{E}_\zeta[H(x,\zeta)]$. The objective $f(x) = g(x) - h(x)$ itself is a difference-of-convex function.

We consider the above stochastic DC program in the online fashion: at each iteration, new data is used to construct a stochastic approximation of G and one of a subgradient of H, and previous samples are no longer available. Since the solution of the proposed algorithms is updated via DCA with proximal terms and the update steps require new fresh samples from the distribution of ξ and ζ, we refer to our algorithm as osproxDCA. The proposed osproxDCA enjoys the double benefit of an online algorithm: it is suitable to perform streaming data from an unknown distribution. Therefore, even if the distribution of ξ and ζ changes at a certain time (suppose that, due to some real-world events, ξ becomes ξ' or ζ becomes ζ' at the k-th iteration), the algorithm will automatically solve the problem with the updated ξ' and ζ'.

We summarize the contribution of the paper as follows:

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- We derive a novel $O(\sqrt{p/n})$ convergence rate (modulo logarithmic factors) in expectation of the point-wise SAA error for subdifferential mappings, matching the same rate of differential mappings in the smooth case, complementing the state-of-art $O(\sqrt[4]{p/n})$ result, where p is the dimension and n is the sample size. Compared to the previous work [29, 14] under uniform setting, we employed a new and simple proof technique by transforming the one-sided deviation of subdifferential set-valued functions into the SAA form of support functions.

Table 1: Convergence rate of the SAA for subdifferential mappings

Paper	Objective setting	Convergence rate
Davis and Drusvyatskiy [14]	Weakly convex, nonsmooth	Uniform, $O(\sqrt[4]{d/n})$
Ruan [29]	Convex-smooth composite with subexponential assumption for random vector and process [29, Assumption C]	Uniform, $O(\sqrt{d/n})$
Ours	Weakly convex, nonsmooth	Point-wise, $O(\sqrt{d/n})$

- Based on this, we propose an adaptive online stochastic proxDCA framework complementing previous work [17, 19]. Under the same setting as [19], our work improves either the sampling scale or the assumptions required.

Table 2: Online stochastic DCA: Sample size at iteration k

Method	Assumption		Sample size	
	Convex part	Concave part	Convex part	Concave part
osDCA [19]	Nonsmooth	Nonsmooth	$O(k^2)$	$O(k^2)$
	Nonsmooth	Smooth	$O(k^2)$	$O(k)$
Ours	Nonsmooth	Nonsmooth	$O(k^2)$	$O(k)$

- From the perspective of practical numerical experiments, our adaptive strategy improves the algorithm's performance and reduces its worst-case sampling size.

1.1 Related Literature

Sample Average Approximation (SAA) and Convergence of Subdifferential Mappings The most general asymptotic convergence result that we are aware of is given in [30]. This study proves that, with probability one, the limit of an enlarged empirical subdifferential converges to the enlarged subdifferential of the population risk under the Hausdorff metric. In the uniform setting, [29] achieves a tight $\sqrt{p/n}$ rate (modulo logarithmic factors) for the uniform convergence of subdifferential mappings, where p is the dimension and n is the sample size. This result is based on a weaker notion of subdifferential convergence and assumes stronger conditions [29, Assumption C]. This complements the $\sqrt[4]{p/n}$ rate in [14], which is based on the graphical distance between subdifferentials, derived using Attouch's epigraphical convergence theorem [2] and Moreau envelope smoothing [24].

Our work advances the literature on the convergence of subdifferential mappings in the context of Sample Average Approximation. We achieve the $O(\sqrt{p/n})$ rate, as found in [29], but without relying on the assumption [29, Assumption C]. Moreover, our result surpasses the $O(\sqrt[4]{p/n})$ rate in [14], offering a new perspective for establishing convergence of subdifferential mappings. Specifically, our approach applies to stochastic weakly-convex classes, which are central to many data-driven applications [13].

Stochastic and Online DC Optimization DC programming, first introduced by [35], has been the subject of extensive study over the years. A thorough review of DC programming's evolution can be found in [20]. While there has been considerable work on DC programming, its application in stochastic settings has only recently gained attention. The majority of existing research on stochastic DC optimization builds on the classical DC Algorithm (DCA) used in deterministic DC problems. The central idea behind DCA is to approximate the DC problem as a convex problem by linearizing the second component. This results in a double-loop algorithm, where the update rule is given by $\min_x \{\phi(x) - \langle \nabla\phi(x_k), x\rangle\}$. The method was first extended to stochastic DC problems (SDCA) in [36] for large-sum problems involving non-convex smooth functions, which was later generalized to handle large-sum non-smooth problems in [18]. The first work that

allowed both components in a DC problem to be non-smooth was presented in [17], where a SDCA scheme was proposed that stores all past information for constructing future subproblems. This approach achieves near-optimal sample complexity by adding just one sample per DCA subproblem. In [19], the efficiency of the SDCA scheme was improved by eliminating the need to store historical information, instead resampling multiple samples at each DCA subproblem and using Sample Average Approximation (SAA) to approximate the linearized DC function. The resulting algorithms exhibit adaptive capabilities, a significant advantage over those proposed in [17]. However, the approach in [19] relies on strong assumptions, as outlined in [19, Assumption 1]. Our work is conducted under similar conditions to [19], improving upon the sampling scale and the assumptions required.

Moreover, non-asymptotic convergence analysis for stochastic DC optimization has been explored in [26, 40]. These studies propose stochastic proximal DC algorithms (SPD), which modify SDCA by adding an extra quadratic term after linearizing the second component. The main limitation of these analyses is their reliance on the smoothness or Hölder continuity of the gradient, assumptions that are often too strong for many commonly used non-smooth functions. Recent work in non-smooth weakly-convex optimization [12, 32, 25, 41] introduced Moreau envelope smoothing approximations for both components, leading to non-asymptotic convergence analysis to nearly ϵ-critical points under deterministic settings. However, these works have yet to establish non-asymptotic convergence for non-smooth DC problems, as a gap remains between nearly ϵ-critical points and true critical points.

Adaptive Sampling in Stochastic Optimization

2 Background and notations

We first summarize the notation used throughout the paper. We write $\mathbb{R}^p$ as the p-dimensional Euclidean space equipped with the inner product $\langle x, y\rangle = x^\top y$ and the induced norm $\|x\| \triangleq \sqrt{x^\top x}$. The symbol $\mathbb{B}(x, \delta)$ is used to denote the closed ball of radius $\delta > 0$ centered at a vector $x \in \mathbb{R}^p$. Let A and C be two nonempty subsets of $\mathbb{R}^n$. The distance from a vector $x \in \mathbb{R}^n$ to A is defined as $\operatorname{dist}(x, A) \triangleq \inf_{y\in A} \|y - x\|$. The one-sided deviation of A from C is defined as $\mathbb{D}(A, C) \triangleq \sup_{x\in A} \operatorname{dist}(x, C)$. The Hausdorff distance between A and C is defined as $\mathbb{H}(A, C) := \max\{\mathbb{D}(A, C), \mathbb{D}(C, A)\}$.

Next, we introduce the concepts of **generalized derivatives and subdifferentials for nonsmooth functions**. Interested readers are referred to the monographs [10, 28, 23] for thorough discussions on these subjects. Consider a function $r : \mathcal{O} \to \mathbb{R}$ defined on an open set $\mathcal{O} \subseteq \mathbb{R}^p$. The classical one-sided directional derivative of r at $\bar{x} \in \mathcal{O}$ along the direction $d \in \mathbb{R}^p$ is defined as $r'(\bar{x}; d) \triangleq \lim_{t\downarrow 0} \dfrac{r(\bar{x} + td) - r(\bar{x})}{t}$ if this limit exists. The function r is said to be directionally differentiable at $\bar{x} \in \mathcal{O}$ if it is directionally differentiable along any direction $d \in \mathbb{R}^p$. In contrast, the Clarke directional derivative of r at $\bar{x} \in \mathcal{O}$ along the direction $d \in \mathbb{R}^p$ is defined as $r^\circ(\bar{x}; d) \triangleq \limsup_{x\to\bar{x},\, t\downarrow 0} \dfrac{r(x + td) - r(x)}{t}$, which is finite when r is Lipschitz continuous near $\bar{x}$. The Clarke subdifferential of r at $\bar{x}$ is the set $\partial_C r(\bar{x}) \triangleq \{v \in \mathbb{R}^n \mid r^\circ(\bar{x}; d) \geq v^\top d \text{ for all } d \in \mathbb{R}^p\}$, which coincides with the usual subdifferential in convex analysis for a convex function. If r is strictly differentiable at $\bar{x}$, then $\partial_C r(\bar{x}) = \{\nabla r(\bar{x})\}$. We say that r is Clarke regular at $\bar{x} \in \mathcal{O}$ if r is directionally differentiable at $\bar{x}$ and $r^\circ(\bar{x}; d) = r'(\bar{x}; d)$ for all $d \in \mathbb{R}^p$. This Clarke regularity at $\bar{x}$ is equivalent to have $r(x) \geq r(\bar{x}) + \bar{v}^\top(x - \bar{x}) + o(\|x - \bar{x}\|)$ for any $\bar{v} \in \partial_C r(\bar{x})$. Therefore, if a function fails to satisfy the Clarke regularity at $\bar{x}$, there does not exist an approximate linear lower bound of the original function based on the Clarke subdifferentials with a small o error locally.

Then, we introduce the concepts of **outer semicontinuity**. Let $\mathcal{A} : \mathbb{R}^p \rightrightarrows \mathbb{R}^m$ be a set-valued mapping. Its outer limit at $x \in \mathbb{R}^n$ is defined as $\limsup_{x\to\bar{x}} \mathcal{A}(x) := \bigcup_{x^\nu\to\bar{x}} \limsup_{\nu\to\infty} \mathcal{A}(x^\nu) = \big\{u \mid \exists\, x^\nu \to \bar{x}, \exists\, u^\nu \to u \text{ with } u^\nu \in \mathcal{A}(x^\nu)\big\}$. We say $\mathcal{A}$ is outer semicontinuous (osc) at $\bar{x} \in \mathbb{R}^n$ if $\limsup_{x\to\bar{x}} \mathcal{A}(x) \subseteq \mathcal{A}(\bar{x})$. Clarke subdifferential is outer semicontinuous, which is necessary for establishing subsequential convergence [28, Proposition 6.6]. In addition, for a L_r-Lipschitz r, $\partial r(\bar{x})$ is locally bounded by L_r [28, Theorem 9.13]. For any $\varepsilon > 0$, we denote the Goldstein subdifferential as $\partial^\varepsilon r(x) := \bigcup_{x'\in\mathbb{B}(x,\varepsilon)} \partial r(x')$., which is also outer semicontinuous [16],

We next review some basics about **random set-valued mappings and their expectation**. Let $(\Omega, \mathcal{F}, P)$ be a probability space. For fixed x, let $\mathcal{A}(x, \omega) : \Omega \to 2^{\mathbb{R}^n}$ be a set-valued mapping which takes values of

closed subsets in $\mathbb{R}^n$. Let $\mathfrak{B}(\mathbb{R}^n)$ or simply $\mathfrak{B}$ denote the space of closed bounded subsets of $\mathbb{R}^n$ endowed with topology $\tau_{\mathbb{H}}$ generated by the Hausdorff distance $\mathbb{H}$. We consider the Borel σ-field $\mathcal{G}(\mathfrak{B}, \tau_{\mathbb{H}})$ generated by the $\tau_{\mathbb{H}}$-open subsets of $\mathfrak{B}$. A set-valued mapping $\mathcal{A}(x,\omega) : \Omega \to 2^{\mathbb{R}^n}$ is said to be $\mathcal{F}$-measurable if, for every member $\mathcal{W}$ of $\mathcal{G}(\mathfrak{B},\ \tau_{\mathbb{H}})$, one has $\mathcal{A}^{-1}(\mathcal{W}) \in \mathcal{F}$. By a measurable selection of $\mathcal{A}(x,\omega)$, we refer to a vector $A(x,\omega) \in \mathcal{A}(x,\omega)$, which is measurable. Note that such measurable selections exist if $\mathcal{A}(x,\omega)$ is measurable, see [1] and references therein. For a general set-valued mapping which is not necessarily measurable, the expectation of $\mathcal{A}(x,\omega)$, denoted by $\mathbb{E}[\mathcal{A}(x,\omega)]$, is defined as the collection of $\mathbb{E}[A(x,\omega)]$ where $A(x,\omega)$ is an integrable selection. The integrability is in the sense of Aumann [3]. $\mathbb{E}[\mathcal{A}(x,\omega)]$ is regarded as well defined if $\mathbb{E}[\mathcal{A}(x,\omega)] \in \mathfrak{B}$ is nonempty. A sufficient condition of the well-definedness of the expectation is that $\mathcal{A}(x,\omega)$ is measurable and $\mathbb{E}[\|\mathcal{A}(x,\omega)\|] := \mathbb{E}[\mathbb{H}(0,\mathcal{A}(x,\omega))] < \infty$, see [3, Theorem 2]. Let $r(x,\xi) : \mathbb{R}^n \times \Xi \to \mathbb{R}$ be a random lsc function, where $\xi : (\Omega, \mathcal{F}, P) \to \Xi$ is a random vector with support set $\Xi \subset \mathbb{R}^m$. Suppose: (a) r is Lipschitz w.r.t. x for every ξ with modulus $\kappa(\xi)$, where $\kappa : \Xi \to \mathbb{R}_+$ is a measurable function that satisfies $\mathbb{E}[\kappa(\xi(\omega))] < \infty$, (b) for any $x \in \mathbb{R}^n$, $r(x,\xi)$ is Clarke regular at x for a.e. $\xi \in \Xi$. Then at every $x \in \mathbb{R}^n$, $\mathbb{E}[r(x,\xi(\omega))]$ is also Clarke regular and $\partial_x \mathbb{E}[r(x,\xi(\omega))] = \mathbb{E}\left[\partial_x r(x,\xi(\omega))\right]$ [10, Theorem 2.7.2].

Finally, we introduce **DC criticality**. A point $x^* \in \mathbb{R}^p$ is called *a DC critical* point if $0 \in \partial g(x^*) - \partial h(x^*)$, or equivalently $\partial g(x^*) \cap \partial h(x^*) \neq \emptyset$, while it is called *a strongly DC critical point* of $(\mathcal{P})$ if $\emptyset \neq \partial h(x^*) \subset \partial g(x^*)$. In the framework of DC programming, the terminology "critical point" is referred to the notion of DC criticality. The notion of DC criticality is close to Clarke, Fréchet, limiting stationarity in the sense $\partial^F f(x^*) \subseteq \partial f(x^*) \subseteq \partial g(x^*) - \partial h(x^*)$ (whenever h is continuous at x^*) and $\partial^C f(x^*) \subseteq \partial g(x^*) - \partial h(x^*)$, where equalities hold under technical assumptions. Furthermore, the commonly used *directional stationarity* is equivalent to strong DC criticality, which is the strongest necessary condition for local DC optimality.

3 osproxDCA framework and SAA error analysis

We first make the following assumptions on $G(x,\xi)$ and $H(x,\zeta)$:

Assumption 1. *(Assumptions For Problem)*

1. *The feasible region C is convex and close, and $f(x) > f_m$ is lower bounded.*
2. *For almost all $\xi \in \Omega_1$, $G(x,\xi)$ is ρ_g-convex (possibly weakly convex if $\rho_g < 0$) over the feasible region C, i.e., for all $x, y \in C$ and $\alpha \in [0,1]$, we have:*

$$G(\alpha x + (1-\alpha)y, \xi) \leq \alpha G(x,\xi) + (1-\alpha)G(y,\xi) - \frac{\rho_g}{2}\alpha(1-\alpha)\|x-y\|^2.$$

   Similarly, $H(x,\zeta)$ is ρ_h-convex (possibly weakly convex ? if $\rho_h < 0$) over C for almost all $\zeta \in \Omega_2$.
3. *For almost all $\xi \in \Omega_1$, $G(x,\xi)$ is L_g-Lipschitz continuous over C, i.e., for all $x, y \in C$, we have:*

$$\|G(x,\xi) - G(y,\xi)\| \leq L_g\|x-y\|.$$

   Similarly, $H(x,\zeta)$ is L_h-Lipschitz continuous over C for almost all $\zeta \in \Omega_2$.

3.1 osproxDCA framework

The DC algorithm is an iterative method with a locally approximated convex objective. At iteration k and iterate x_k, the local approximation model to the objective with the exact g and a true subgradient $y_k \in \partial h(x_k)$ is

$$m_k(x) = g(x) - h(x_k) - y_k^T(x - x_k) + \frac{1}{2}\mu_k\|x - x_k\|^2, \tag{1}$$

where $\mu_k > 0$ is a scalar quadratic coefficient. In the stochastic setting, we use stochastic estimate $\bar{g}_k(\cdot)$ of $g(\cdot)$ and a stochastic subgradient estimate $\bar{y}_k$ of y_k in the local model $\bar{m}_k(\cdot)$:

$$\bar{m}_k(x) = \bar{g}_k(x) - \bar{h}_k(x_k) - \bar{y}_k^T(x - x_k) + \frac{1}{2}\mu_k\|x - x_k\|^2. \tag{2}$$

The stochastic estimate of function value $\bar{h}_k$ does not affect the optimization subproblem and its solution. Hence, we hereby use it interchangeably with $h(x_k)$. Denoting $d = x - x_k$, $\bar{m}_k(x)$ and $m_k(x)$ can be rewritten

as $\bar{M}_k(d)$ and $M_k(d)$, respectively,

$$\begin{aligned}\bar{M}_k(d) &= \bar{g}_k(x_k+d) - \bar{h}_k(x_k) - \bar{y}_k^T d + \frac{1}{2}\mu_k\|d\|^2,\\ M_k(d) &= g(x_k+d) - h(x_k) - y_k^T d + \frac{1}{2}\mu_k\|d\|^2.\end{aligned} \tag{3}$$

The subproblem to be solved at iteration k is

$$\begin{aligned}\underset{d}{\text{minimize}} \quad & \bar{M}_k(d)\\ \text{subject to} \quad & x_k + d \in C.\end{aligned} \tag{4}$$

The first-order optimality conditions of subproblem (4) are

$$\begin{aligned}\bar{z}_{k+1} - \bar{y}_k + \mu_k \bar{d}_k + \bar{v}_k &= 0,\\ x_k, x_k + \bar{d}_k \in C, \bar{v}_k &\in \partial i_C(x_k + \bar{d}_k),\end{aligned} \tag{5}$$

where $\bar{z}_{k+1} \in \partial \bar{g}_k(x_k + \bar{d}_k)$.

The stochastic quantities are estimated at each iteration with independent and identically distributed (i.i.d.) general samples of ξ and ζ. We use $S_{g,k}$ to denote the set of samples of ξ at iteration k when obtaining estimation $g(x)$, and $S_{h,k}$ to denote the set of samples of ζ at iteration k when obtaining estimation $\bar{y}_k$. The cardinality or sample size of $S_{g,k}, S_{h,k}$ is defined as $N_{g,k} = |S_{g,k}|$ and $N_{h,k} = |S_{h,k}|$. Let $\mathcal{F}_k \triangleq \sigma\left(S_{g,1}, S_{h,1}, S_{g,2}, S_{h,2}, \ldots, S_{g,k-1}, S_{h,k-1}\right)$ be a filtration, i.e., an increasing sequence of σ-fields generated by samples used in the previous $k-1$ iterations.

Our proposed stochastic DC algorithm framework is presented in Algorithm 1, while the exact rule when updating $\mu_k, N_{g,k}, N_{h,k}$ is analyzed and presented in Section 4 and 5. Additionally, the quadratic term $\mu_k\|d\|^2$ can be replaced by $d^T B_k d$, where B_k is a positive definite matrix.

Algorithm 1 osproxDCA framework

1: Initialize $x_0, \mu_0, N_{g,0}, N_{h,0}$.
2: **for** $k = 0, 1, 2, \cdots$ **do**
3: Generate iid samples $S_{g,k} = \{\xi^{k,i}\}_{i=1}^{N_{g,k}}$ and $S_{h,k} = \{\zeta^{k,i}\}_{i=1}^{N_{h,k}}$ from the distribution of ξ and ζ that are independent of previous samples.
4: Construct the local approximation model $\bar{M}_k(d)$ by SAA:
Set $\bar{g}_k(x) = \frac{1}{N_{g,k}}\sum_{i=1}^{N_{g,k}} G\left(x, \xi^{k,i}\right)$, $\bar{h}_k(x) = \frac{1}{N_{h,k}}\sum_{i=1}^{N_{h,k}} H\left(x, \zeta^{k,i}\right)$, and select $\bar{y}_k \in \partial\bar{h}_k(x) = \frac{1}{N_{h,k}}\sum_{i=1}^{N_{h,k}} \partial_x H\left(x, \zeta^{k,i}\right)$.
5: Solve the $\rho_g + \mu_k$-convex optimization subproblem (4) to obtain $\bar{d}_k$.
6: Take the step $x_{k+1} = x_k + \bar{d}_k$.
7: Call the $\mu_k, N_{g,k}, N_{h,k}$ update rules to obtain $\mu_{k+1}, N_{g,k+1}, N_{h,k+1}$.
8: **end for**

It is trivial to verify that $\bar{g}_k(x)$ is L_g-Lipschitz, $\bar{h}_k(x)$ is L_h-Lipschitz, and $\|\bar{y}_k\| \le L_h$.

Remark 1. *If there exists an isomorphic mapping ϕ from $(\Omega_1, \mathcal{F}_1, \mathbb{P}_1)$ to $(\Omega_2, \mathcal{F}_2, \mathbb{P}_2)$, Step 3 in Algorithm 1 could be simplified to:*

If $N_{g,k} \ge N_{h,k}$,

1. *generate iid samples $S_{g,k} = \{\xi^{k,i}\}_{i=1}^{N_{g,k}}$ from the distribution of ξ that are independent of previous samples;*
2. *for $i = 1, 2, \cdots, N_{h,k}$, let $\zeta^{k,i} = \phi(\xi^{k,i})$ and $S_{h,k} = \{\zeta^{k,i}\}_{i=1}^{N_{h,k}}$.*

The counterpart is similar.

3.2 Uniform convergence of the SAA for g in expectation

To continue, we first introduce a lemma regarding the convergence rate of the sample average approximation in expectation. This result is derived from the Rademacher average of the random function $\psi(x;\omega)$, as

discussed in [15, Corollary 3.2] and further explored in [11, Theorem 10.1.5]. For a random vector function $\psi(\cdot,\omega):\mathcal{X}\to\mathbb{R}$ and i.i.d. random variables $(\omega^1,\ldots,\omega^n)=\bar{\omega}^n$ drawn from the distribution of ω, we define the quantity

$$\delta_n\left(\psi,\bar{\omega}^n\right):=\sup_{x\in\mathcal{X}}\left\|\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{\psi}\left(x,\omega^i\right)-\mathbb{E}\boldsymbol{\psi}(x,\omega)\right\|.$$

We have the following basic estimates [15, Theorem 3.1].

Lemma 1. *(Basic Estimates). If functions $\psi(\cdot,\omega)$ are bounded by constant M and Lipschitz continuous with constant L in the first variable $x\in\mathcal{X}\subseteq[0,r]^p\subseteq\mathbb{R}^d$ uniformly in ω, then for any $\alpha\in(0,1/2)$ it holds that*

$$\begin{aligned}&\mathbb{E}_{\bar{\omega}^n}\delta_n\left(\psi,\bar{\omega}^n\right)\leq 2\sqrt{p}(Lr+M/\sqrt{(1-2\alpha)e})/n^{\alpha},\\&\mathbb{P}\left\{\sqrt{n}\left|\delta_n\left(\psi,\bar{\omega}^n\right)-\mathbb{E}_{\bar{\omega}^n}\delta_n\left(\psi,\bar{\omega}^n\right)\right|\geq t\right\}<2\exp\left\{-t^2/\left(2M^2\right)\right\}\quad\forall t\geq 0.\end{aligned}$$

By Lemma 1, we could derive the SAA error in expectation for g.

Theorem 1. *Assume that $\bar{d}_k\in[-r_k,r_k]^d$, for any $\alpha\in(0,1/2)$,*

$$\mathbb{E}_{S_{g,k}}\left|\bar{g}_k(x_{k+1})-\bar{g}(x_k)-g(x_{k+1})+g_k(x_k)\right|\leq\frac{C_g}{\mu_k N_{g,k}^{\alpha}},$$

where $C_g\geq 4\sqrt{p}L_g(L_g+L_h)\left(2+\frac{L_g}{\sqrt{(1-2\alpha)e}}\right)$.

Proof. First we prove that $\|\bar{d}_k\|\leq\frac{2(L_g+L_h)}{\mu_k}$. Since $\bar{d}_k$ is the solution of subproblem (4), we have

$$\bar{g}_k(x_k+\bar{d}_k)-\bar{h}_k(x_k)-\bar{y}_k^T\bar{d}_k+\frac{1}{2}\mu_k\|\bar{d}_k\|^2=\bar{M}_k(\bar{d}_k)\leq\bar{M}_k(0)=\bar{g}_k(x_k)-\bar{h}_k(x_k).$$

Hence we have $\frac{1}{2}\mu_k\|\bar{d}_k\|^2\leq\bar{g}_k(x_k)-\bar{g}_k(x_k+\bar{d}_k)+\bar{y}_k^T\bar{d}_k\leq\|\bar{d}_k\|(L_g+L_h)$, which implies $\|\bar{d}_k\|\leq\frac{2(L_g+L_h)}{\mu_k}$.

Let $r_k=\frac{2(L_g+L_h)}{\mu_k}$ and $\psi(x,\xi)=G(x,\xi)-G(x_k,\xi)$, which is L_g-Lipschitz. By Lemma 1,

$$\mathbb{E}_{Sg,k}\sup_{\Delta_x\in[-r_k,r_k]^d}\left|\frac{1}{n}\sum_{i=1}^{n}\psi\left(x_k+\Delta_x,\xi^{k,i}\right)-\mathbb{E}_{\xi}\psi(x_k+\Delta_x,\xi)\right|\leq\frac{\hat{C}_g}{N_{g,k}^{\alpha}},$$

where $\hat{C}_g=2\sqrt{p}(2L_gr_k+L_gr_k/\sqrt{(1-2\alpha)e})$. Noticed that

$$\left|\bar{g}_k(x_{k+1})-\bar{g}(x_k)-g(x_k)+g_k(x_k)\right|\leq\sup_{\Delta_x\in[-r_k,r_k]^d}\left|\frac{1}{n}\sum_{i=1}^{n}\psi\left(x_k+\Delta_x,\xi^{k,i}\right)-\mathbb{E}_{\xi}\psi(x_k+\Delta_x,\xi)\right|,$$

Hence, we finish the proof by substituting r_k for its definition. □

3.3 Convergence of the SAA for subdifferential mapping ∂h in expectation

In the following discussion, we investigate the convergence rate of the sample average subdifferential mapping $\partial\bar{h}_k(x)=\frac{1}{N_{k,h}}\sum_{k=1}^{N}\partial_x h\left(x,z^k\right)$ to $\partial h(x)=\mathbb{E}_{\zeta}\left[\partial_x h(x,\zeta)\right]$ in expectation. To achieve this, we analyze the quantity $\mathbb{D}\left(\frac{1}{N}\sum_{k=1}^{N}\partial_x H\left(x,\zeta^k\right),\mathbb{E}_{\zeta}\left[\partial_x H(x,\zeta)\right]\right)$ converging to 0 in expectation as N increases.

Define the support function of a set S as $\sigma(u,S):=\sup_{s\in S}u^Ts$. It is well known that $\sigma(u,S)=\sigma(u,\operatorname{conv}S)$ and for nonempty sets S and S', the following holds:

$$\sigma\left(u,S+S'\right)=\sigma(u,S)+\sigma\left(u,S'\right),\tag{6}$$

as discussed in [9, p. 621]. Furthermore, the well-established Hömander's formula [8, Theorem II-18] states that for any two nonempty convex and compact subsets A and B of $\mathbb{R}^n$:

$$\mathbb{D}(A,B)=\max_{\|u\|\leqslant 1}\left(\sigma(u,A)-\sigma(u,B)\right).\tag{7}$$

Using the above formula, we have the following lemma that converts our targeted quantity into the form of an SAA error; see, e.g., [39, Theorem 4.3].

Lemma 2. *For any $x \in C$, we have*

$$\mathbb{D}\left(\partial \bar{h}_k(x), \partial h(x)\right) = \max_{\|u\| \leqslant 1} \left[\frac{1}{N_{k,h}} \sum_{i=1}^{N_{k,h}} \sigma\left(u, \partial_x H\left(x, \zeta^{k,i}\right)\right) - \mathbb{E}_\zeta\left[\sigma\left(u, \partial_x H(x,\zeta)\right)\right] \right].$$

Proof.

$$\begin{aligned}
&\mathbb{D}\left(\frac{1}{N_{h,k}} \sum_{i=1}^{N_{h,k}} \partial_x H\left(x, \zeta^k\right), \operatorname{conv} \mathbb{E}_\zeta\left[\partial_x H(x,\zeta)\right] \right) \\
&= \mathbb{D}\left(\operatorname{conv}\left\{ \frac{1}{N_{h,k}} \sum_{i=1}^{N_{h,k}} \partial_x H\left(x, \zeta^{k,i}\right) \right\}, \operatorname{conv} \mathbb{E}_\zeta\left[\partial_x H(x,\zeta)\right] \right) \\
&= \sup_{\|u\| \leqslant 1} \sigma\left(u, \frac{1}{N_{h,k}} \sum_{i=1}^{N_{h,k}} \partial_x H\left(x, \zeta^{k,i}\right) \right) - \sigma\left(u, \mathbb{E}_\zeta\left[\partial_x H(x,\zeta)\right]\right) \\
&= \sup_{\|u\| \leqslant 1} \left[\frac{1}{N_{h,k}} \sum_{i=1}^{N_{h,k}} \sigma\left(u, \partial_x H\left(x, \zeta^{k,i}\right)\right) - \sigma\left(u, \mathbb{E}_\zeta\left[\partial_x H(x,\zeta)\right]\right) \right] \\
&= \sup_{\|u\| \leqslant 1} \left[\frac{1}{N_{h,k}} \sum_{i=1}^{N_{h,k}} \sigma\left(u, \partial_x H\left(x, \zeta^{k,i}\right)\right) - \mathbb{E}_\zeta\left[\sigma\left(u, \partial_x H(x,\zeta)\right)\right] \right].
\end{aligned}$$

By the convexity of $H(x,\zeta)$, $\partial_x H\left(x, \zeta^k\right)$ is convex and compact, hence $\frac{1}{N_{h,k}} \sum_{i=1}^{N_{h,k}} \partial_x H\left(x, \zeta^{k,i}\right)$ is convex and the first equality stands. The second equality is due to (7). The third equality is due to (6). The last equality is due to the interchangeability of $\mathbb{E}_\zeta$ and σ, see [27, Proposition 3.4] for details. □

We now derive the subdifferentials convergence of the SAA for h in expectation:
Lemma 3. *For any $\alpha \in (0, 1/2)$, we have*

$$\sup_{x \in C} \mathbb{E}_{S_{h,k}}\left[\mathbb{D}\left(\partial \bar{h}_k(x), \partial h(x)\right)\right] \leq \frac{\hat{C}_h}{N_{h,k}^{\alpha}},$$

where $\hat{C}_h = 2\sqrt{p}(2L_h + L_h/\sqrt{(1-2\alpha)e})$.

Moreover,

$$\mathbb{P}\left\{ N_{h,k}^{\alpha_h} \mathbb{D}\left(\partial \bar{h}_k(x), \partial h(x)\right) \geq \hat{C}_h + t \right\} \leq \exp\left\{ -t^2 / \left(2L_h^2\right) \right\}.$$

Proof. For any $x \in C$, by Lemma 2,

$$\mathbb{E}_{S_{h,k}}\left[\mathbb{D}\left(\partial \bar{h}_k(x), \partial h(x)\right)\right] \leq \mathbb{E}_{S_{h,k}} \sup_{\|u\| \leqslant 1} \left\| \frac{1}{N_{k,h}} \sum_{i=1}^{N_{k,h}} \psi(u, \zeta^{k,i}) - \mathbb{E}_\zeta\left[\psi(u,\zeta)\right] \right\|,$$

where $\psi(u,\zeta) = \sigma\left(u, \partial_x h\left(x,\zeta\right)\right)$. To satisfy the condition in Lemma 1, we first verify that $\psi(\cdot,\zeta)$ are uniformly bounded by constant L_h and Lipschitz continuous with constant L_h in the first variable $u \in \mathbb{B}(0,1) \subseteq [-1,1]^d$ uniformly in ζ.

The first property is trivial since $\sup_{s \in \partial_x h(x,\zeta)} \|s\| \leq L_h$. Now, we prove the second property. For any $u, v \in \mathbb{B}(0,1)$, suppose that $\sigma\left(u, \partial_x h\left(x,\zeta\right)\right) = u^T s$ where $s \in \partial_x h\left(x,\zeta\right)$. Then we have

$$\sigma\left(v, \partial_x h\left(x,\zeta\right)\right) \geq v^T s \geq u^T s - \|u-v\|\|s\| \geq \sigma\left(u, \partial_x h\left(x,\zeta\right)\right) - L_h\|u-v\|.$$

Similarly, $\sigma\left(u, \partial_x h\left(x,\zeta\right)\right) \geq \sigma\left(v, \partial_x h\left(x,\zeta\right)\right) - L_h\|u-v\|$, hence $\psi(u,\zeta)$ is L_h-Lipschitz continuous in $\mathbb{B}(0,1)$.

We thus finish the proof by using Lemma 1. □

We then strengthen the result to bound the square error in expectation as follows. This is the main result of this section.

Theorem 2. *For any $\alpha \in (0, 1/2)$, we have*

$$\sup_{x\in C} \mathbb{E}_{S_{h,k}} \left[\mathbb{D}^2 \left(\partial \bar{h}_k(x), \partial_x h(x)\right)\right] \leq \frac{C_h}{N_{h,k}^{2\alpha}},$$

where $C_h \geq \hat{C}_h \left(\hat{C}_h + \sqrt{2}L_h \frac{\sqrt{\alpha'}}{2\sqrt{(\alpha'-\alpha)e}}\right) + L_h^2$ *with* $\hat{C}_h = \sqrt{p}(2L_h + L_h/\sqrt{(1-2\alpha')e})$ *for any* $\alpha' \in (\alpha, 1/2)$.

Proof. For any $x \in C$, let $\delta = \mathbb{D}\left(\partial \bar{h}_k(x), \partial_x h(x)\right)$, which is bounded by L_h. By Lemma 3, $\mathbb{E}_{S_{h,k}}[\delta] \leq \frac{\hat{C}_h}{N_{h,k}^{\alpha}}$ and $\mathbb{P}\left\{\delta \geq \frac{\hat{C}_h + t}{N_{h,k}^{\alpha_h}}\right\} \leq \exp\left\{-t^2/\left(2L_h^2\right)\right\}$ stands for any $t > 0$, $\alpha' \in (\alpha, 1/2)$.

Hence for any $t > 0$, we have

$$\begin{aligned}
\mathbb{E}_{S_{h,k}}[\delta^2] &= \mathbb{E}_{S_{h,k}}\left[\delta^2 \mathbf{1}(\delta < \frac{\hat{C}_h + t}{N_{h,k}^{\alpha_h}})\right] + \mathbb{E}_{S_{h,k}}\left[\delta^2 \mathbf{1}(\delta \geq \frac{\hat{C}_h + t}{N_{h,k}^{\alpha'}})\right] \\
&\leq \mathbb{E}_{S_{h,k}}[\delta] \frac{\hat{C}_h + t}{N_{h,k}^{\alpha'}} + L_h^2 \mathbb{P}\left\{\delta \geq \frac{\hat{C}_h + t}{N_{h,k}^{\alpha'}}\right\} \\
&\leq \frac{\hat{C}_h(\hat{C}_h + t)}{N_{h,k}^{2\alpha'}} + L_h^2 \exp\left\{-t^2/\left(2L_h^2\right)\right\},
\end{aligned}$$

where the first inequality is due to $\delta \leq L_h$, and the last inequality is due to Lemma 3.

Take $t = \sqrt{2}L_h\sqrt{\alpha' \ln N_{h,k}}$, we could obtain that

$$\mathbb{E}_{S_{h,k}}[\delta^2] \leq \frac{\hat{C}_h\left(\hat{C}_h + \sqrt{2}L_h\sqrt{\alpha' \ln N_{h,k}}\right) + L_h^2}{N_{h,k}^{2\alpha'}}.$$

Since $\sqrt{\ln N_{h,k}} \leq \frac{N^{2\alpha'-2\alpha}}{2\sqrt{(\alpha'-\alpha)e}}$, we have for any $\alpha' \in (\alpha, 1/2)$,

$$\mathbb{E}_{S_{h,k}}[\delta^2] \leq \frac{\hat{C}_h\left(\hat{C}_h + \sqrt{2}L_h \frac{\sqrt{\alpha'}}{2\sqrt{(\alpha'-\alpha)e}}\right) + L_h^2}{N_{h,k}^{2\alpha}},$$

where $\hat{C}_h = \sqrt{p}(2L_h + L_h/\sqrt{(1-2\alpha')e})$. □

Remark 2. *Notice that the second concentration-type result of Lemma 1 and 3 are due to McDiarmid's bounded difference inequality, and they play an essential role in the proof of Theorem 2.*

Remark 3. *When H is smooth, Theorem 2 could be simplified to*

$$\sup_{x\in C} \mathbb{E}_{S_{h,k}} \left[\|\nabla \bar{h}_k(x) - \nabla h(x)\|^2\right] \leq \frac{L_h^2}{N_{h,k}},$$

that is, $C_h = L_h^2$ and $\alpha = 1/2$.

4 Convergence analysis

4.1 Descent and error analysis per iteration

This section analyzes the descent and error at the k-th iteration. We could derive the following sufficiently descent property, similar to the original deterministic version, see, e.g., [33, Theorem 3] and [34, Theorem 3.7].

Lemma 4. *(Sufficiently Descent Property) For any $y_k \in \partial h(x_k)$, the step x_{k+1} from Algorithm 1 satisfies*

$$f(x_k) - f(x_{k+1}) \geq (y_k - \bar{y}_k)^T \bar{d}_k + \mu_{\rho,k} \left\|\bar{d}_k\right\|^2 + \Delta g_k(x_k) - \Delta g_k(x_{k+1}),$$

where $\mu_{\rho,k} = \mu_k + \frac{\rho_g+\rho_h}{2}$ and $\Delta g_k(x) = g(x) - \bar{g}_k(x)$.

Proof. From the convexity of $\bar{g}_k(\cdot)$ and $h(\cdot)$, we have

$$\begin{aligned} &\bar{g}_k(x_k) - \bar{g}_k(x_k + \bar{d}_k) + \bar{z}_{k+1}^T \bar{d}_k \geq \frac{1}{2}\rho_g \|\bar{d}_k\|^2, \\ &h(x_k + \bar{d}_k) - h(x_k) - y_k^T \bar{d}_k \geq \frac{1}{2}\rho_h \|\bar{d}_k\|^2, \end{aligned} \tag{8}$$

for $\bar{z}_{k+1} \in \bar{\partial}\bar{g}_k(x_k + \bar{d}_k)$, $x_k + \bar{d}_k \in C$, $d_k \in \mathbb{R}^n$. Therefore,

$$\begin{aligned} &f(x_k) - f(x_{k+1}) = g(x_k) - h(x_k) - g(x_{k+1}) + h(x_{k+1}) \\ =&\bar{g}_k(x_k) - \bar{g}_k(x_{k+1}) - h(x_k) + h(x_{k+1}) + [g(x_k) - \bar{g}_k(x_k)] - [g(x_{k+1}) - \bar{g}_k(x_{k+1})] \\ \geq& - \bar{z}_{k+1}^T \bar{d}_k + y_k^T \bar{d}_k + \frac{1}{2}(\rho_g + \rho_h)\|\bar{d}_k\|^2 + \Delta g_k(x_k) - \Delta g_k(x_{k+1}). \end{aligned} \tag{9}$$

Taking the dot product with $-\bar{d}_k$ of the first line of (5), we have

$$\begin{aligned} -\bar{z}_{k+1}^T \bar{d}_k + \bar{y}_k^T \bar{d}_k =& \mu_k \|\bar{d}_k\|^2 + \bar{v}_k \bar{d}_k \\ \geq& \mu_k \|\bar{d}_k\|^2 + [i_C(x_k) - i_C(x_k + \bar{d}_k) - \bar{v}_k^T(-\bar{d}_k)] \\ \geq& \mu_k \|\bar{d}_k\|^2, \end{aligned} \tag{10}$$

where the first inequality uses the second line of (5) and the second inequality comes from the convexity of C and $i_C(\cdot)$. Applying (10) to (9) leads to

$$f(x_k) - f(x_{k+1}) \geq (y_k - \bar{y}_k)^T \bar{d}_k + \mu_{\rho,k}\|\bar{d}_k\|^2 + \Delta g_k(x_k) - \Delta g_k(x_{k+1}). \tag{11}$$

□

Remark 4. *In practice, experience from the deterministic problems suggests that μ_k be reduced when possible to improve convergence behavior. For details, see section 3.1 of [38].*

To further the analysis, the estimation error bound in expectation from sampling, which is derived in Section 3, comes into play. In [21], with no specific form of $\bar{\phi}_k(\cdot)$, the variance of the stochastic objectives is assumed to be bounded. The variance bounds then come from sample size N_k. Similarly, in [5], an unbiased estimate with bounded variance is made on the gradient with SQP. In [31], the Monte Carlo estimate of the objective is assumed to be unbiased, and its variance is uniformly bounded.

We present the sufficient descent property in expectation as follows.

Lemma 5. *(Sufficiently Descent Property in Expectation) At the k-th iteration, the following stands for any $c > 0$:*

$$\mathbb{E}\left[f(x_k) - f(x_{k+1}) \,|\mathcal{F}_k\right] \geq (\mu_{\rho,k} - c)\mathbb{E}\left[\left\|\bar{d}_k\right\|^2 |\mathcal{F}_k\right] - \frac{C_g}{\mu_k N_{g,k}^{\alpha_g}} - \frac{C_h}{4cN_{h,k}^{\alpha_h}}, \tag{12}$$

where $\alpha_g \in (0, 1/2)$, $\alpha_h \in (0, 1]$ with corresponding C_g, C_h are defined in Theorem 1 and 2.

Proof. By Lemma 4, for any $y_k \in \partial h(x_k)$, $c > 0$,

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\geq (y_k - \bar{y}_k)^T \bar{d}_k + \mu_{\rho,k}\left\|\bar{d}_k\right\|^2 + \Delta g_k(x_k) - \Delta g_k(x_{k+1}) \\ &\geq (\mu_{\rho,k} - c)\left\|\bar{d}_k\right\|^2 + \Delta g_k(x_k) - \Delta g_k(x_{k+1}) - \frac{1}{4c}\|y_k - \bar{y}_k\|^2. \end{aligned}$$

Take y_k such that $dist(\bar{y}_k, \partial h(x_k)) = \|y_k - \bar{y}_k\|$, we have

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\geq (\mu_{\rho,k} - c)\left\|\bar{d}_k\right\|^2 + \Delta g_k(x_k) - \Delta g_k(x_{k+1}) - \frac{1}{4c} dist(\bar{y}_k, \partial h(x_k))^2 \\ &\geq (\mu_{\rho,k} - c)\left\|\bar{d}_k\right\|^2 + \Delta g_k(x_k) - \Delta g_k(x_{k+1}) - \frac{1}{4c}\mathbb{D}^2\left(\partial\bar{h}_k(x_k), \partial_x h(x_k)\right) \end{aligned}$$

Taking expectation for both sides under $\mathcal{F}_k$, we finish the proof after Theorem 1 and Theorem 2. □

4.2 Almost sure convergence to critical points

From Lemma 5, to obtain a convergent step $\bar{d}_k \to 0$, the sequences $N_{g,k}$ and $N_{h,k}$ need to be controlled so that $\sum_{k\geq 0}\left(\frac{1}{\mu_k N_{g,k}^{\alpha_g}}+\frac{1}{N_{h,k}^{\alpha_h}}\right)<\infty$. In order to obtain a finite $N_{g,k}$ at each iteration, we make the necessary assumption that μ_k is bounded below by a constant greater than 0.

Assumption 2. *μ_k is bounded below by $\check{\mu}>0$.*

Under Assumption 2, there exist a number of combinations of conditions to ensure such is the case. For example, this can be achieved as standard practice by increasing the sample size $N_{g,k}$ and $N_{h,k}$ as the following condition:

Condition 1. ***(Summable Condition)*** *Given pre-determined sequences $\{\hat{N}_{g,k}\}$ and $\{\hat{N}_{h,k}\}$ satisfy $\sum_{k\geq 0}\left(\hat{N}_{h,k}^{-\alpha_h}+\hat{N}_{g,k}^{-\alpha_g}\right)<\infty$ and a pre-determined constant $c_l, c_\mu>0$, we say that* ***Summable Condition*** *is satisfied at the k-th iteration if parameters $c_k, \mu_k, N_{g,k}, N_{h,k}$ are selected to satisfy*

$$N_{g,k}\geq \hat{N}_{g,k},\ N_{h,k}\geq \hat{N}_{h,k},\ \text{and } c_l\leq c_k\leq \mu_{\rho,k}-c_\mu. \tag{13}$$

We could ensure the convergence of osproxDCA by directly setting a sublinear growth for the sample size. However, this approach is not adaptive. We propose a practical **Stepsize Norm Condition** for adaptive sampling in the following part. Compared to the gradient accuracy condition and other variance-based tests in [**?**, **?**], our method bridges the gap between theory and practical implementation.

Condition 2. ***(Stepsize Norm Condition)*** *Given a pre-determined constant $c_\mu>0$, we say that* ***Stepsize Norm Condition*** *is satisfied at the k-th iteration if parameters $c_k, \mu_k, N_{g,k}, N_{h,k}$ are selected to satisfy*

$$\left(\mu_{\rho,k-1}-c_\mu-c_{k-1}\right)\|\bar{d}_{k-1}\|^2\geq \frac{C_g}{\mu_k N_{g,k}^{\alpha_g}}+\frac{C_h}{4c_k N_{h,k}^{\alpha_h}},\ \text{and } c_k\leq \mu_{\rho,k}-c_\mu. \tag{14}$$

Remark 5. *Here, $\{c_k\}$ are intermediate variables for parameter updates. They provide the interrelationship between μ_k, $N_{g,k}$, and $N_{h,k}$ to ensure the convergence of osproxDCA in an expectation. Note that it is only used to determine μ_k, $N_{g,k}$, and $N_{h,k}$, and is irrelevant to the solution of the subproblems.*

We present the convergence result in the following theorems, which are significant for designing practical adaptive osproxDCA.

Theorem 3. *(Square Summable Property) Under Assumption 2, if either* ***Summable Condition*** (13) *or* ***Stepsize Norm Condition*** (14) *is satisfied for sufficiently large k, then we have*

$$\lim_{k\to\infty}\mathbb{E}\left[\sum_{k\geq 0}\left\|\bar{d}_k\right\|^2|\mathcal{F}_0\right]<\infty\ ,\ \text{hence } \mathbb{E}\left[\left\|\bar{d}_k\right\||\mathcal{F}_0\right]\to 0 \text{ as } k\to\infty.$$

Further, $\lim_{k\to\infty}\|\bar{d}_k\|=0$ with probability 1.

Proof. Let $\mathcal{K}_1$ be the set of k that satisfies Summable Condition, $\mathcal{K}_2$ be the set of k that satisfies Stepsize Norm Condition. Supposed that when $k\geq K$, $k\in\mathcal{K}_1\cup\mathcal{K}_2$. If $k\in\mathcal{K}_2$, we have

$$\mathbb{E}\left[f\left(x_k\right)-f\left(x_{k+1}\right)|\mathcal{F}_k\right]\geq\left(\mu_{\rho,k}-c_k\right)\mathbb{E}\left[\left\|\bar{d}_k\right\|^2|\mathcal{F}_k\right]-\left(\mu_{\rho,k-1}-c_\mu-c_k\right)\|\bar{d}_{k-1}\|^2.$$

Taking expectation under $\mathcal{F}_0$ from both side, we have

$$\mathbb{E}\left[f\left(x_k\right)-f\left(x_{k+1}\right)|\mathcal{F}_0\right]\geq\left(\mu_{\rho,k}-c_k\right)\mathbb{E}\left[\left\|\bar{d}_k\right\|^2|\mathcal{F}_0\right]-\left(\mu_{\rho,k-1}-c_\mu-c_k\right)\mathbb{E}\left[\|\bar{d}_{k-1}\|^2|\mathcal{F}_0\right].$$

If $k\in\mathcal{K}_1\cup\{0,1,\cdots,K-1\}$, from Lemma 5 and take expectation under $\mathcal{F}_0$, we have

$$\mathbb{E}\left[f\left(x_k\right)-f\left(x_{k+1}\right)|\mathcal{F}_0\right] \geq (\mu_{\rho,k}-c_k)\mathbb{E}\left[\left\|\bar{d}_k\right\|^2|\mathcal{F}_0\right]-\frac{C_g}{\mu_k N_{g,k}^{\alpha_g}}-\frac{C_h}{4c_k N_{h,k}^{\alpha_h}}.$$

Taking sum for each k, we have

$$\begin{aligned}\mathbb{E}\left[\sum_{k=0}^{n-1}\left(\mu_{\rho,k}-c_k\right)\|\bar{d}_k\|^2|\mathcal{F}_0\right] &\leq \mathbb{E}\left[f\left(x_0\right)-f\left(x_n\right)|\mathcal{F}_0\right]+\sum_{k=0}^{K-1}\left(\frac{C_g}{\mu_k N_{g,k}^{\alpha_g}}+\frac{C_h}{4c_k N_{h,k}^{\alpha_h}}\right)\\ &+\sum_{k=K,k\in\mathcal{K}_1}^{n-1}\left(\frac{C_g}{\mu_k N_{g,k}^{\alpha_g}}+\frac{C_h}{4c_k N_{h,k}^{\alpha_h}}\right)+\mathbb{E}\left[\sum_{k=K,k\in\mathcal{K}_2}^{n-1}\left(\mu_{\rho,k-1}-c_{k-1}-c_\mu\right)\|\bar{d}_{k-1}\|^2|\mathcal{F}_0\right].\end{aligned} \tag{15}$$

Hence, we could obtain that

$$\begin{aligned}\mathbb{E}\left[\sum_{k=K-1}^{n-1}c_\mu\|\bar{d}_k\|^2|\mathcal{F}_0\right] &\leq \mathbb{E}\left[f\left(x_0\right)-f\left(x_n\right)|\mathcal{F}_0\right]\\ &+\sum_{k=0}^{K-1}\left(\frac{C_g}{\mu_k N_{g,k}^{\alpha_g}}+\frac{C_h}{4c_k N_{h,k}^{\alpha_h}}\right)+\sum_{k=K,k\in\mathcal{K}_1}^{n-1}\left(\frac{C_g}{\mu_k N_{g,k}^{\alpha_g}}+\frac{C_h}{4c_l N_{h,k}^{\alpha_h}}\right).\end{aligned} \tag{16}$$

By Assumption 1, $\mathbb{E}\left[f\left(x_0\right)-f\left(x_n\right)|\mathcal{F}_0\right] < f(x_0)-f_m$. By the definition of Summable Condition and Assumption 2, we have

$$\sum_{k=K,k\in\mathcal{K}_1}^{n-1}\left(\frac{C_g}{\mu_k N_{g,k}^{\alpha_g}}+\frac{C_h}{4c_l N_{h,k}^{\alpha_h}}\right) \leq \sum_{k=K,k\in\mathcal{K}_1}^{n-1}\left(\frac{C_g}{\check{\mu}\hat{N}_{g,k}^{\alpha_g}}+\frac{C_h}{4c_l \hat{N}_{h,k}^{\alpha_h}}\right).$$

Take $n\to\infty$, the right-hand side of (16) is bounded. It follows that $\lim_{n\to\infty}\mathbb{E}[\|\bar{d}_n\|^2|\mathcal{F}_0]=0$. The first part of the theorem is proven.

We proceed by contradiction for the second part of the theorem. Suppose there exists $\epsilon>0$ and $a>0$ such that

$$\mathbb{P}(\limsup_{k\to\infty}\|\bar{d}_k\|\geq\epsilon)\geq a. \tag{17}$$

By Chebyshev's inequality, we have $\mathbb{P}(\|\bar{d}_k\|\geq\epsilon)\leq\frac{\mathbb{E}\left[\|\bar{d}_k\|^2|\mathcal{F}_0\right]}{\epsilon^2}$. Since $\mathbb{E}\left[\|\bar{d}_k\|^2|\mathcal{F}_0\right]$ is finitely summable, there exists $N>0$ such that $\sum_{k=N}^{\infty}\mathbb{P}(\|\bar{d}_k\|\geq\epsilon)\leq\sum_{k=N}^{\infty}\frac{\mathbb{E}\left[\|\bar{d}_k\|^2|\mathcal{F}_0\right]}{\epsilon^2}<a$. Therefore,

$$\mathbb{P}\left(\limsup_{k\to\infty}\|\bar{d}_k\|\geq\epsilon\right)=\mathbb{P}\left(\limsup_{k\to\infty:k\geq N}\|\bar{d}_k\|\geq\epsilon\right)\leq\sum_{k=N}^{\infty}\mathbb{P}(\|\bar{d}_k\|\geq\epsilon)<a. \tag{18}$$

This is a contradiction against (17). Hence, we could obtain that $\lim_{k\to\infty}\|\bar{d}_k\|=0$ with probability 1. □

In the next part, we prove the convergence to critical points. The following assumption is natural and necessary; without it, the algorithm may not converge to critical points, even in the deterministic case.

Assumption 3. *μ_k is bounded above by $\hat{\mu}>0$.*

To proceed, we first prove a technical lemma.

Lemma 6. *Under Assumption 3, if either* ***Summable Condition*** (13) *or* ***Stepsize Norm Condition*** (14) *is satisfied for sufficiently large k, then $N_{g,k}, N_{h,k}\to\infty$ as $k\to\infty$ with probability 1.*

Proof. Supposed that when $k\geq K$, $k\in\mathcal{K}_1\cup\mathcal{K}_2$. If $k\in\mathcal{K}_1$, we have $N_{g,k}\geq\hat{N}_{g,k}$ and $N_{h,k}\geq\hat{N}_{h,k}$. Since $\sum_{k\geq 0}\left(N_{h,k}^{-\alpha_h}+N_{g,k}^{-\alpha_g}\right)<\infty$, we have $N_{g,k}, N_{h,k}\to\infty$ as $k\to\infty$.

If $k \in \mathcal{K}_2$, we have

$$\begin{aligned}\frac{C_g}{\hat{\mu} N_{g,k}^{\alpha_g}} + \frac{C_h}{4(\hat{\mu} + \frac{\rho_g+\rho_h}{2} - c_\mu) N_{h,k}^{\alpha_h}} &\leq \frac{C_g}{\mu_k N_{g,k}^{\alpha_g}} + \frac{C_h}{4(\mu_k + \frac{\rho_g+\rho_h}{2} - c_\mu) N_{h,k}^{\alpha_h}} \\ &\leq \frac{C_g}{\mu_k N_{g,k}^{\alpha_g}} + \frac{C_h}{4 c_k N_{h,k}^{\alpha_h}} \\ &\leq (\mu_{\rho,k-1} - c_\mu - c_{k-1}) \|\bar{d}_{k-1}\|^2 \\ &\leq \left(\hat{\mu} + \frac{\rho_g + \rho_h}{2} - c_\mu - c_{k-1}\right) \|\bar{d}_{k-1}\|^2,\end{aligned}$$

where the first and last inequality is due to Assumption 3. The second and third inequality is due to the definition of Stepsize Norm Condition (14).

By Theorem 3, $\lim_{k\to\infty} \|\bar{d}_k\| = 0$ with probability 1, thus $N_{g,k}, N_{h,k} \to \infty$ with probability 1. □

We then introduce a result concerning the law of large numbers (LLN) for subdifferentials, which follows from the LLN for random set-valued mappings as discussed in [1]. This result can be interpreted as a pointwise version of [30, Theorem 2].

Lemma 7. *For any $r > r' \geqslant 0$ the following limits hold:*

$$\sup_{x\in\mathcal{C}} \mathbb{D}\left(\partial^{r'} \bar{g}_k(x), \partial^r g(x)\right) \to 0 \ \ w.p.\ 1.$$

$$\sup_{x\in\mathcal{C}} \mathbb{D}\left(\partial^{r'} \bar{h}_k(x), \partial^r h(x)\right) \to 0 \ \ w.p.\ 1.$$

Now, we derive our main result as follows.

Theorem 4. *Under Assumption 2 and 3, if either* ***Summable Condition*** (13) *or* ***Stepsize Norm Condition*** (14) *is satisfied for sufficiently large k, every accumulation point of the sequence $\{x_k\}$ produced by Algorithm 1 is a DC critical point of f with probability 1.*

Proof. Let x be an accumulation point of $\{x_k\}$. Firstly we prove that for any given ε, $0 \in \partial^\varepsilon g(x) - \partial^\varepsilon h(x) + \partial i_C(x)$.

From the optimal condition (5), there exist $\bar{z}_{k+1} \in \partial \bar{g}_k(x_{k+1})$ and $\bar{v}_k \in \partial i_C(x_{k+1})$ for each k, such that

$$\bar{z}_{k+1} - \bar{y}_k + \mu_k \bar{d}_k + \bar{v}_k = 0. \tag{19}$$

By Lemma 7, there exists $y_k \in \partial^\varepsilon h(x_k)$, $z_k \in \partial^\varepsilon g(x_{k+1})$ such that $\lim_{k\to\infty} \bar{y}_k - y_k = 0$ and $\lim_{k\to\infty} \bar{z}_k - z_k = 0$ with probability 1. We can assume $\lim_{k\to\infty} x_{n_k} = x$ where $x \in C$. Further, y_k and z_k are bounded due to the Lipschitz continuity of $G(x,\xi)$ and $H(x,\zeta)$. Thus, $\bar{v}_k$ is also bounded, and there exist accumulation points for $\{y_k\}$ and $\{z_k\}$. Passing on to a subsequence, we assume $y_{n_{m_k}} \to \bar{y}$ and $z_{n_{m_k}} \to \bar{z}$. Therefore $\bar{y}_{n_{m_k}} \to \bar{y}$ and $\bar{z}_{n_{m_k}} \to \bar{z}$. By (19), we have

$$0 = \bar{z}_{n_{m_k}} - \bar{y}_{n_{m_k}} + \mu_{n_{m_k}} \bar{d}_{n_{m_k}} + \bar{v}_{n_{m_k}}. \tag{20}$$

By Theorem 3 and Assumption 3, $\lim_{k\to\infty} \mu_k \bar{d}_k = 0$ with probability 1. Thus, $\lim_{k\to\infty} \bar{v}_{n_{m_k}} = \bar{y} - \bar{z}$ with probability 1. Given that $\bar{v}_{n_{m_k}} \in \partial i_C(x_{n_{m_k}+1})$, the outer semicontinuity of $\partial i_C(\cdot)$ leads to $\lim_{k\to\infty} \bar{v}_{n_{m_k}} \in \partial i_C(x)$. By the outer semicontinuity of Goldstein subdifferential, $\bar{z} = \lim_{k\to\infty} z_{n_{m_k}} \in \partial^\varepsilon g(x)$, since $x_{n_{m_k}+1} \to x$ and $z_{n_{m_k}} \in \partial^\varepsilon g(x_{n_{m_k}+1})$. Similarly, $\bar{y} = \lim_{k\to\infty} y_{n_{m_k}} \in \partial h^\varepsilon(x)$.

Therefore, with probability 1, $0 \in \partial^\varepsilon g(x) - \partial^\varepsilon h(x) + \partial i_C(x)$. By the outer semicontinuity of Clarke subdifferential, we obtain that

$$\bigcap_{\epsilon>0} (\partial^\epsilon g(x) - \partial^\epsilon h(x) + \partial i_C(x)) = \partial g(x) - \partial h(x) + \partial i_C(x).$$

Thus we have $0 \in \partial g(x) - \partial h(x) + \partial i_C(x)$ with probability 1. □

Remark 6. *Indeed, the current solution can be considered as the initial point for restart. Hence, even if the distribution of ξ and ζ changes at a certain time, the algorithm continues operating based on new samples from the new distribution. Theorem 3 and 4 are still valid, and the subsequential convergence with probability one to DC critical points of the DC problem associated with the new distribution is guaranteed. This is indeed an advantage of the osproxDCA. In contrast, intuitively, stochastic algorithms using aggregated updates (still using old samples to compute the current solution) barely have this kind of adaptivity.*

5 Adaptive sampling osproxDCA

The previous theorems are enough to get a simple adaptive scheme for osproxDCA, following a similar strategy of [6, 7, 4, 37]. We present the adaptive sampling osproxDCA as Algorithm 2.

Algorithm 2 Adaptive osproxDCA

Require: Initial point x_0, initial parameter μ_0, c_0, $N_{g,0}$ and $N_{h,0}$, error estimation parameter $\alpha_g \in (0, 1/2)$, $\alpha_h \in (0, 1]$ with corresponding C_g, C_h defined in Theorem 1 and 2, step size norm condition parameter $c_\mu, c_l > 0$, sample size upper bound sequence $\{\hat{N}_{g,k}\}$ and $\{\hat{N}_{h,k}\}$ which satisfy $\sum_{k\geq 0}\left(\hat{N}_{h,k}^{-\alpha_h} + \hat{N}_{g,k}^{-\alpha_g}\right) < \infty$, step size upper bound $\hat{\mu}$ and lower bound $\check{\mu}$ that satisfy $\hat{\mu} > \check{\mu}$.

1: **for** $k = 0, 1, 2, \cdots$ **do**

2: Generate iid samples $\{\xi^{k,i}\}_{i=1}^{N_{g,k}}$ and $\{\zeta^{k,i}\}_{i=1}^{N_{h,k}}$ from the distribution of ξ and ζ that are independent of previous samples.

3: Construct the local approximation model $\bar{M}_k(d)$:
Set $\bar{g}_k(x) = \frac{1}{N_{g,k}}\sum_{i=1}^{N_{g,k}} G\left(x, \xi^{k,i}\right)$, $\bar{h}_k(x) = \frac{1}{N_{h,k}}\sum_{i=1}^{N_{h,k}} H\left(x, \zeta^{k,i}\right)$, and select $\bar{y}_k \in \partial\bar{h}_k(x) = \frac{1}{N_{h,k}}\sum_{i=1}^{N_{h,k}} \partial_x H\left(x, \zeta^{k,i}\right)$.

4: Solve the $(\rho_g + \mu_k)$-convex optimization subproblem (4) to obtain $\bar{d}_k$:

$$\begin{aligned} \underset{d}{\text{minimize}} \quad & \bar{g}_k(x_k + d) - \bar{h}(x_k) - \bar{y}_k^T d + \frac{1}{2}\mu_k\|d\|^2, \\ \text{subject to} \quad & x_k + d \in C. \end{aligned}$$

5: Take the step $x_{k+1} = x_k + \bar{d}_k$.

6: Update $\check{\mu} \leq \mu_{k+1} \leq \hat{\mu}$, $c_{k+1}, N_{g,k+1}$ and $N_{h,k+1}$ such that one of the followings stands:
1. $\left(\mu_k + \frac{\rho_g+\rho_h}{2} - c_\mu - c_k\right)\|\bar{d}_k\|^2 \geq \frac{C_g}{\mu_{k+1}N_{g,k+1}^{\alpha_g}} + \frac{C_h}{4c_k N_{h,k+1}^{\alpha_h}}$, and $c_{k+1} \leq \mu_{k+1} + \frac{\rho_g+\rho_h}{2} - c_\mu$,
2. $N_{g,k+1} \geq \hat{N}_{g,k+1}$, $N_{h,k} \geq \hat{N}_{h,k+1}$, and $c_l \leq c_{k+1} \leq \mu_{k+1} + \frac{\rho_g+\rho_h}{2} - c_\mu$.

7: **end for**

When $\rho_g + \rho_h > 0$, given any predetermined sequence $\{\mu_k\}$ that satisfy $0 < \check{\mu} \leq \mu_k \leq \hat{\mu}$, we present a simplified version for Algorithm 2 as follows by fixing $c_k = \frac{\rho_g+\rho_h}{2} = c_l$, $c_\mu = \frac{\check{\mu}}{2}$, see Algorithm 3.

There has now arisen a natural question:

What is the optimal strategy for updating the sample sizes $N_{g,k}$ and $N_{h,k}$?

The following lemma provides the optimal order between $N_{g,k}$ and $N_{h,k}$ to minimize the total number of samples at each iteration.

Lemma 8. *(Optimal Order of N_g and N_h) Consider the following subproblem when updating the sample size:*

$$\begin{aligned} \textit{minimize} \quad & N_{g,k} + N_{h,k}, \\ \textit{subject to} \quad & \frac{C_1}{N_{g,k}^{1/2}} + \frac{C_2}{N_{h,k}} \leq C_0, \\ N_{g,k} \geq 0, \quad & N_{h,k} \geq 0. \end{aligned}$$

We could derive that $N_{h,k} = \sqrt{2}\left(\frac{C_2}{C_1}\right)^{1/2} N_{g,k}^{3/4}$. Hence the optimal order of $N_{h,k}$ is $O(s_{g,k}^{3/4})$.

Remark 7. *If the updating rule of $N_{g,k}$ and $N_{h,k}$ is based on Lemma 8, then sample size upper bound sequence $\hat{N}_{h,k}$ is no longer required.*

Algorithm 3 Simplified Adaptive osproxDCA

Require: Initial point x_0, error estimation parameter $\alpha_g \in (0, 1/2)$, $\alpha_h \in (0, 1]$ with corresponding C_g, C_h defined in Theorem 1 and 2, sample size upper bound sequence $\{\hat{N}_{g,k}\}$ and $\{\hat{N}_{h,k}\}$ which satisfy $\sum_{k\geq 0}\left(\hat{N}_{h,k}^{-\alpha_h} + \hat{N}_{g,k}^{-\alpha_g}\right) < \infty$, predetermined quadratic term sequence $\{\mu_k\}$ with upper bound $\hat{\mu}$ and lower bound $\check{\mu}$.

1: **for** $k = 0, 1, 2, \cdots$ **do**

2: Generate iid samples $\{\xi^{k,i}\}_{i=1}^{N_{g,k}}$ and $\{\zeta^{k,i}\}_{i=1}^{N_{h,k}}$ from the distribution of ξ and ζ that are independent of previous samples.

3: Construct the local approximation model $\bar{M}_k(d)$:

Set $\bar{g}_k(x) = \frac{1}{N_{g,k}}\sum_{i=1}^{N_{g,k}} G\left(x, \xi^{k,i}\right)$, $\bar{h}_k(x) = \frac{1}{N_{h,k}}\sum_{i=1}^{N_{h,k}} H\left(x, \zeta^{k,i}\right)$, and select $\bar{y}_k \in \partial\bar{h}_k(x) = \frac{1}{N_{h,k}}\sum_{i=1}^{N_{h,k}} \partial_x H\left(x, \zeta^{k,i}\right)$.

4: Solve the $\rho_g + \mu_k$-convex optimization subproblem (4) to obtain $\bar{d}_k$:

$$\begin{aligned} \underset{d}{\text{minimize}} \quad & \bar{g}_k(x_k + d) - \bar{h}(x_k) - \bar{y}_k^T d + \frac{1}{2}\mu_k\|d\|^2, \\ \text{subject to} \quad & x_k + d \in C. \end{aligned}$$

5: Take the step $x_{k+1} = x_k + \bar{d}_k$.

6: Update $N_{g,k+1}$ and $N_{h,k+1}$ such that one of the followings stands:

1. $\left(\mu_k - \frac{\check{\mu}}{2}\right)\|\bar{d}_k\|^2 \geq \frac{C_g}{\mu_{k+1}N_{g,k+1}^{\alpha_g}} + \frac{C_h}{2(\rho_g+\rho_h)N_{h,k+1}^{\alpha_h}}$,
2. $N_{g,k+1} \geq \hat{N}_{g,k+1}$, and $N_{h,k} \geq \hat{N}_{h,k+1}$.

7: **end for**

6 Numerical applications

6.1 Objective with smooth concave part

Consider the expected problem of principal component analysis (denoted by (E-PCA)) [22],

$$\min\left\{f(x) = -(1/2)\mathbb{E}_{s\sim\mathbb{P}}\left(\langle x, s\rangle^2\right) : x \in \mathbb{R}^p\|x\| \leq 1\right\},$$

where s is a normalized random vector, i.e., $\|s\| = 1$. We formulate the (E-PCA) problem as an SDC program with $G(x, s) = \frac{L}{2}\|x\|^2 - \frac{1}{2}\langle x, s\rangle^2, H(x, s) = \frac{L}{2}\|x\|^2 + \frac{1}{2}\langle x, s\rangle^2$, feasible region $C = \mathbb{B}(0, 1)$, where $L \geq 1$. Then for all s, $G(\cdot, s)$ is Lipschitz with modulus L and convex with modulus $L - 1$ on C. $H(\cdot, s)$ is Lipschitz with modulus $L + 1$ and convex with modulus L on C. In Algorithm 3, we take constant quadratic term $\mu_k = \mu$, $\alpha_g = 0.45$ with corresponding $C_g = \sqrt{p}8L(L + 1)(2L + 1)$. Since $H(\cdot, s)$ is smooth, we take $\alpha_h = 1$ with corresponding $C_h = (L + 1)^2$.

6.2 Objective with nonsmooth concave part

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